



Git Guide.

Every Git and GitHub answer in one place.

Ask in plain language, or paste the error Git printed.
Exact commands, a danger level for each, and its undo.

an undo for everything · works offline · no trackers · MIT



Amey Thakur

amey-thakur.github.io/GIT-GUIDE



Every Git and GitHub answer in one place.

A note from Amey

Git carries nearly every codebase on earth, and almost everyone learned it by accident: a command from a teammate, a midnight search, a ritual repeated without understanding.

Git Guide exists to end that way of learning. Ask in plain language, or paste the error Git printed, and get the exact commands with the two things most resources never give: how dangerous each one is, and how to take it back.

These pages are the portable half: the model, the full command reference, an undo for every situation, and the errors you will meet. The other half is a search box, where a thousand answers and five hundred decoded errors wait at **amey-thakur.github.io/GIT-GUIDE**.

Amey



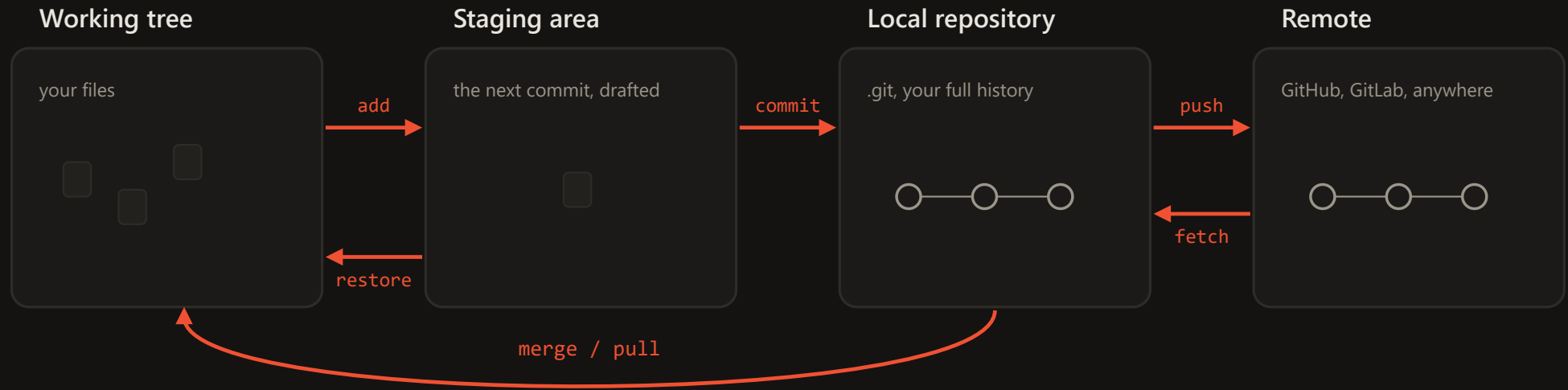
Amey Thakur

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Where your code lives



add copies a change into the staging area, the draft of your next commit. You choose what goes in.

push uploads your commits; **fetch** downloads theirs and stops. **pull** is fetch plus merge.

commit seals the draft into a permanent snapshot in your local repository. Still only on your machine.

restore copies recorded versions back over your files; **reset** moves the branch pointer itself. That is where history is un-happened.



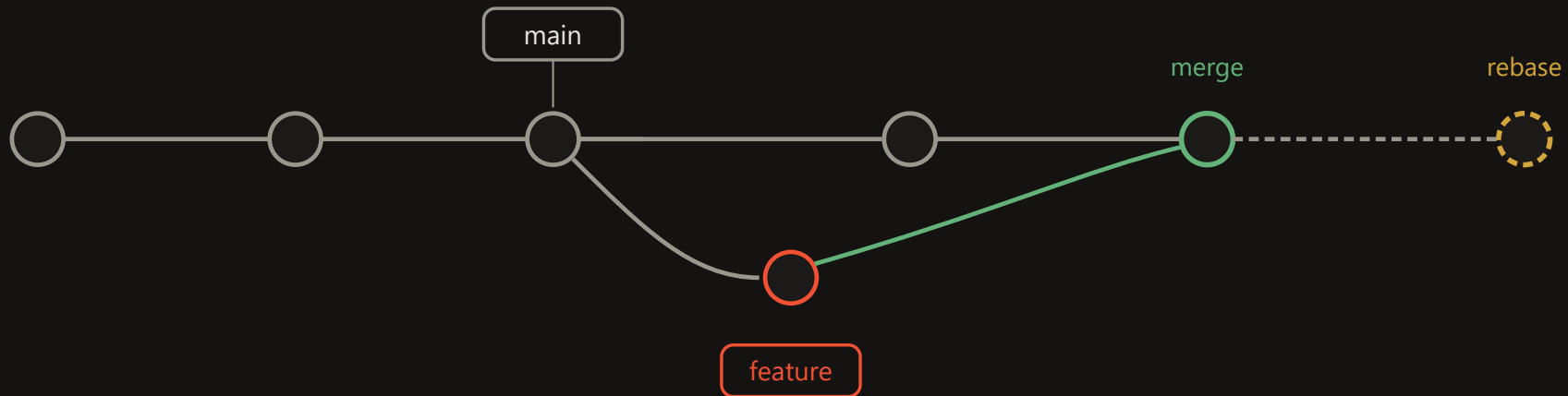
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Every Git and GitHub answer in one place.

Commits, branches, HEAD



A **branch** is a movable label on one commit. Creating one copies nothing; it is instant and free.

A **rebase** replays your commits as new ones for a straight line. It rewrites history: your own unshared work only.

A **merge** commit has two parents and joins the lines. History shows what truly happened; always safe on shared branches.

HEAD is where you stand, normally attached to a branch. Standing on a commit directly is detached HEAD: a state, not an error.



From zero to first push

1 Install

Windows: winget install Git.Git · macOS: brew install git · Debian: sudo apt install git

```
git --version
```

3 One key, every host

Add the public key to GitHub, GitLab, Bitbucket, Azure: the same key works on all

```
ssh-keygen -t ed25519 -C "<email>"
```

5 The loop you will run forever

Edit, stage, commit, push. Everything else is a variation

```
git add <file> && git commit -m "<message>" && git push
```

2 Identity

The same email as your GitHub account, or commits will not count on your profile

```
git config --global user.name "<name>" && git config --global user.email "<email>"
```

4 First repository

Joining: git clone <url> · Publishing a folder of yours:

```
gh repo create <name> --public --source . --push
```

6 When it goes wrong

It will, for everyone. Ask the site in plain language, or paste the exact error

```
amey-thakur.github.io/GIT-GUIDE
```



Setup and identity

Once per machine; everything else builds on it.

```
git config --global user.name "<name>"
```

Name stamped into your commits

```
git config --global user.email "<email>"
```

Use your host account email so commits link to your profile

```
git config --global init.defaultBranch main
```

New repositories start on main

```
git config --global core.editor "code --wait"
```

Commit messages open in VS Code

```
git config --global pull.ff only
```

Pull never creates surprise merge commits

```
git config --global fetch.prune true
```

Deleted remote branches disappear locally too

```
git config --global rebase.autoStash true
```

Pull with rebase works even with edits in progress

```
git config --global core.autocrlf true
```

Windows line endings handled once; macOS and Linux use input

```
git config --list --show-origin
```

Every setting and the file it comes from

Start a project

From nothing, or from a URL.

```
git init -b main
```

Turn this folder into a repository

```
git clone <url>
```

Copy a repository, any host, full history

```
git clone --depth 1 <url>
```

Latest snapshot only, fastest

```
git remote add origin <url>
```

Connect a local repository to a host

```
git push -u origin main
```

First push; -u links the branches for good



The daily loop

The eight commands that are most of Git.

```
git status -sb
```

What changed, one line per file

```
git diff
```

Unstaged edits, line by line

```
git diff --staged
```

What the next commit will contain

```
git add <file>
```

Stage a change; git add -p picks piece by piece

```
git commit -m "<message>"
```

Seal the staged snapshot

```
git commit --amend --no-edit
```

Add staged changes to the last commit

```
git fetch
```

See what the remote has before touching anything

```
git pull --rebase
```

Get remote commits, replay yours on top

```
git push
```

Publish your commits

Branches

Cheap, fast, and safe to make by the dozen.

```
git switch -c <branch>
```

Create and move in one step

```
git switch <branch>
```

Move to an existing branch

```
git switch -
```

Back to the branch you just left

```
git branch --show-current
```

Where am I

```
git branch -vv
```

Every local branch with its remote and ahead-behind state

```
git branch -a
```

Every branch, local and remote

```
git push -u origin <branch>
```

Publish a new branch

```
git branch -d <branch>
```

Delete a merged branch; -D forces

```
git push origin --delete <branch>
```

Delete it on the remote too

```
git branch -m <old> <new>
```

Rename



Merge and rebase

Two ways to bring lines together.

```
git merge <branch>
```

Bring a branch in; history stays true

```
git merge --no-ff <branch>
```

Always keep the feature visible as one bubble

```
git rebase main
```

Replay your branch on top of main

```
git rebase -i HEAD~<n>
```

Rewrite, reorder, squash your last n commits

```
git cherry-pick <hash>
```

Copy one commit onto this branch

```
git merge --abort
```

Back out of a conflicted merge

```
git rebase --continue
```

Resume after resolving a conflict

Undo and rescue

Every mistake has a way back.

```
git restore <file>
```

Discard edits, back to last commit

```
git restore --staged <file>
```

Unstage, keep the edits

```
git commit --amend -m "<new message>"
```

Rewrite the last commit message

```
git reset --soft HEAD~1
```

Uncommit, keep everything staged

```
git reset --hard HEAD~1
```

Erase the last commit and its work

```
git revert <hash>
```

Undo by adding an opposite commit; safe when pushed

```
git reflog
```

Every state you have been in; the safety net

```
git reset --hard ORIG_HEAD
```

Undo the merge, rebase, or pull that just happened

```
git clean -nd
```

Preview deleting untracked files; -fd deletes



Stash

A shelf for work that is not ready.

```
git stash push -u -m "<label>"
```

Set everything aside, untracked included

```
git stash list
```

What is shelved

```
git stash pop
```

Bring the newest back and drop it

```
git stash apply stash@{<n>}
```

Bring a specific one back, keep it shelved

```
git stash show -p
```

See inside before applying

Inspect and search

Read history like a book with an index.

```
git log --oneline --graph --all
```

The whole history as a graph

```
git log -p -- <file>
```

Every change to one file; --follow crosses renames

```
git log --grep="<text>"
```

Find commits by message

```
git log -S "<text>"
```

Find commits that added or removed this text

```
git diff main...<branch>
```

What the branch changed since it split; the pull request view

```
git shortlog -sn --all
```

Commits per author, ranked

```
git blame -w <file>
```

Who last touched each line, ignoring whitespace

```
git show <hash>
```

One commit in full; --stat for files only

```
git bisect start
```

Binary-search history for the commit that broke it

```
git branch -a --contains <hash>
```

Which branches carry this commit



Tags and releases

Permanent names on permanent commits.

```
git tag -a v1.0.0 -m "<note>"
```

Mark a release

```
git push origin v1.0.0
```

Tags do not push themselves

```
git tag -d <tag>
```

Delete locally; push origin --delete <tag> for the remote

```
git describe --tags
```

Nearest release to where you stand

```
git switch --detach v1.0.0
```

Visit a release; switch - to come back

Files and ignoring

What Git tracks, and what it must never.

```
printf "node_modules/\n" >> .gitignore
```

Ignore by pattern; affects untracked files only

```
git rm --cached <file>
```

Stop tracking, keep on disk

```
git mv <old> <new>
```

Rename and stage in one step

```
git update-index --skip-worktree <file>
```

Ignore your local edits to a tracked file

```
printf "* text=auto\n" >> .gitattributes
```

Line endings settled for the whole team, committed once

```
touch <folder>/.gitkeep
```

Keep an empty folder



Every platform, same Git

Git is identical everywhere; only the remote URL and sign-in differ. The same SSH public key works on every host.

```
git clone git@github.com:<user>/<repo>.git
```

GitHub over SSH

```
git clone git@gitlab.com:<user>/<repo>.git
```

GitLab

```
git clone git@bitbucket.org:<workspace>/<repo>.git
```

Bitbucket

```
git clone https://dev.azure.com/<org>/<project>/_git/<repo>
```

Azure DevOps (Azure Repos)

```
git clone https://git-codecommit.  
<region>.amazonaws.com/v1/repos/<repo>
```

AWS CodeCommit

```
git config --global credential.helper manager
```

Browser sign-in for HTTPS remotes, all hosts

```
git remote set-url origin <url>
```

Move between hosts or between HTTPS and SSH

```
git clone --mirror <old> && git push --mirror <new>
```

Migrate anywhere with full history

No host at all

Git predates the hosts and needs none of them.

```
git init --bare ~/server/repo.git
```

Your own remote on any machine or shared drive

```
git bundle create repo.bundle --all
```

The whole repository as one file: USB, email, air gap

```
git clone repo.bundle -b main <folder>
```

Clone from the file, no network

```
git format-patch -3
```

Last three commits as mailable patch files

```
git am <file>.patch
```

Apply mailed patches, authorship intact

```
git archive --format=zip -o src.zip HEAD
```

Clean source export, no .git



History surgery

Everything here rewrites history. Fresh clone first, backup kept, force-with-lease after, teammates re-clone.

```
git commit --fixup <hash>
```

Mark a fix as belonging to an earlier commit

```
git rebase -i --autosquash <hash>^
```

Melt fixups into their targets

```
git filter-repo --invert-paths --path <file>
```

Erase a file from all history

```
git filter-repo --strip-blobs-bigger-than 10M
```

Erase every giant blob

```
git push --force-with-lease
```

The only force push worth typing

Scale and speed

Big repositories, kept quick.

```
git clone --filter=blob:none --sparse <url>
```

Huge repository, download almost nothing

```
git sparse-checkout set <folders>
```

Materialize only what you work on

```
git worktree add ../<repo>-review <branch>
```

Second folder, second branch, zero stashing

```
git lfs track "*.psd"
```

Large binaries stored properly

```
git count-objects -vH
```

How heavy is this repository really

```
git maintenance start
```

Background upkeep keeps big repositories fast



Submodules

A repository pinned inside another.

```
git submodule add <url> <path>
```

Pin another repository at an exact commit inside yours

```
git clone --recurse-submodules <url>
```

Clone a repo and its submodules in one step

```
git submodule update --init --recursive
```

Fill the empty folders after a plain clone

```
git submodule update --remote
```

Move pins to each submodule's latest commit

Quality of life

Small settings, daily comfort.

```
git config --global help.autocorrect prompt
```

git status asks: run git status instead?

```
git config --global alias.st "status -sb"
```

git st forever

```
git config --global alias.lg "log --oneline --graph --all"
```

git lg: the graph

```
git config --global rerere.enabled true
```

Never resolve the same conflict twice

```
git config --global gpg.format ssh
```

First of three lines to signed, Verified commits

```
git commit --allow-empty -m "rerun"
```

Retrigger CI without touching code

```
git config core.hooksPath .githubhooks
```

Versioned hooks the whole team shares



How teams run Git

> Working alone

Commit at every working state; push is your backup; branch before experiments

```
git add -A && git commit -m "<message>" && git push
```

> Fork

For repositories you cannot push to, which is all of open source

```
gh repo fork <owner>/<repo> --clone
```

> Releases and hotfixes

A release is a tag; a hotfix is a branch from that tag, merged back after shipping

```
git switch -c hotfix-2.1.1 v2.1.0
```

> Feature branch

The team default: main stays releasable, every change arrives by pull request

```
git switch -c <change> && git push -u origin <change>
```

> Trunk-based

Branches live hours, not weeks; tiny changes merge daily behind flags

```
git config --global pull.rebase true
```

The rules that keep teams safe. Never rewrite a branch others build on; rewrite your own freely with force-with-lease. Pull with rebase. Protect main so nothing lands without review and green checks. When anything is unclear: git status names the state you are in, and usually the way out.



Your first repository

Name

Short, lowercase, hyphens: weather-app

Add a README

The front page, and your first commit

Add .gitignore

Pick your language; noise stays out from day one

Choose a license

MIT if unsure; none means all rights reserved

The pull request path

branch · commit · push · open the PR · checks run · review answers with commits · merge, and the branch retires.

`gh pr create --fill`

Open it from the terminal

`gh pr checkout <number>`

Review one locally

`gh pr merge --squash --delete-branch`

Land it clean

The words nobody explains

repository a project and its entire history; everyone says repo

README.md the front page, rendered automatically

Markdown plain text with light marks; GitHub renders it everywhere

.gitignore what Git must never track

LICENSE the legal terms of reuse

public, private everyone reads, only you write; or invitation only

star a bookmark and a thank-you; stars rank repositories

watch subscribe to issues, PRs, and releases

clone vs fork to your machine; to your account

issue a tracked conversation about one bug or request

default branch usually main: what visitors see, what PRs target

gist one shareable file with its own URL

organization a shared account for a team



Undo anything

The situation, the command, and how much it costs. Green is safe, amber rewrites history, red can lose work.

Committed too early	<code>git reset --soft HEAD~1</code>	Safe
Staged the wrong file	<code>git restore --staged <file></code>	Safe
Want the file back as committed	<code>git restore <file></code>	Can lose work
Forgot a file in the last commit	<code>git add <file></code>	Safe
Wrong commit message	<code>git commit --amend -m "<new message>"</code>	Safe
Amended and regretted it	<code>git reset --soft HEAD@{1}</code>	Safe
Merged the wrong branch	<code>git merge --abort</code>	Safe
Rebase went wrong	<code>git reset --hard ORIG_HEAD</code>	Can lose work
Pushed something you should not have	<code>git revert <hash></code>	Safe



Undo anything, continued

Pull brought in a mess	<code>git reset --hard ORIG_HEAD</code>	Can lose work
Need the project back at an old commit	<code>git switch --detach <hash></code>	Safe
One file should go back in time	<code>git restore --source=<hash> <file></code>	Can lose work
A commit vanished	<code>git reflog</code>	Safe
Deleted a branch by mistake	<code>git branch <name> <hash></code>	Safe
Committed on the wrong branch	<code>git branch <new-branch></code>	Rewrites history
A file must leave the last commit	<code>git reset HEAD~1 -- <file></code>	Rewrites history
Need a clean tree for five minutes	<code>git stash push -u -m "<label>"</code>	Safe
Made a repository by accident	<code>rm -rf .git</code>	Can lose work



I lost work: the order to try

Work down this list. Stop at the step that finds it.

1 **git status**

Name the state you are in. Half of all panics end here.

3 **git reflog**

Every position HEAD has held, kept about ninety days. Anything ever committed is here.

5 **git fsck --lost-found**

For work that was staged but never committed: it survives as a dangling object.

2 **git stash push -u**

Freeze everything before trying anything, so no fix can destroy more.

4 **git branch rescue <hash>**

Found it? Name it immediately. A named commit cannot be garbage collected.

6 **Editor local history**

Never staged, never committed? Git never saw it. Your editor may have.

The habit that makes this unnecessary. Commit early and often, and push daily. The reflog protects anything committed; nothing protects work that was never committed.



How much does this command cost

Every answer in this guide carries one of these three marks, and its own undo.

Safe

Nothing is lost, and most of it is reversible in a keystroke

`status` · `log` · `diff` · `fetch` · `switch` · `stash` · `revert` · `tag`

Rewrites history

The content survives, but commits change identity. Fine alone, rude on a shared branch

`rebase` · `commit --amend` · `reset --soft` · `filter-repo` · `push --force-with-lease`

Can lose work

Git may hold no copy afterwards. Rehearse first, and know the undo before you press enter

`reset --hard` · `clean -fd` · `push --force` · `branch -D` · `checkout -- <file>`

Two commands hold nothing back. `git clean` deletes files Git never tracked, and `git reset --hard` discards edits that were never committed. Everything else, the reflog can usually reach.



The errors you will actually meet

The message, the cause, the first command. All 500 decoded errors are searchable **on the site**.

fatal: not a git repository (or any of the parent directories):

.git

This folder, and none above it, contains a .git directory. You are outside the project.

cd <project-folder>

git@github.com: Permission denied (publickey).

The host never received a key it recognizes: no key loaded, key not added to your account, or an HTTPS problem disguised by an SSH URL.

ssh -T git@github.com

fatal: Authentication failed for 'https://github.com/...'

The credential your machine sent is wrong, expired, or a password where a token is required.

gh auth login

! [rejected] main -> main (non-fast-forward)

The remote has commits you do not have. Git refuses to overwrite them.

git pull --rebase

! [rejected] main -> main (fetch first)

Same cause as non-fast-forward: someone pushed before you.

git pull --rebase

CONFLICT (content): Merge conflict in <file>

Both sides changed the same lines; Git needs a human decision.

git status

fatal: Unable to create '.git/index.lock': File exists.

Another Git process is running, or one crashed and left its lock behind.

rm -f .git/index.lock

You are in 'detached HEAD' state.

Not an error. You checked out a commit or tag directly, so you stand on a snapshot instead of a branch.

git switch -c <name>



Errors, continued

Your branch and 'origin/main' have diverged

Both you and the remote gained different commits since you last synced.

```
git pull --rebase
```

error: src refspec main does not match any

The branch you are pushing does not exist locally, almost always because there is no commit yet.

```
git status
```

fatal: refusing to merge unrelated histories

The two sides share no common commit, typically a fresh git init pulled against a remote that already has commits.

```
git pull origin main --allow-unrelated-histories
```

fatal: detected dubious ownership in repository

The repository folder belongs to a different OS user than the one running Git, common in containers, WSL, and network drives.

```
git config --global --add safe.directory <path>
```

remote: error: File <name> is <size> MB; this exceeds GitHub's file size limit of 100.00 MB

One specific file crossed the hard limit; the push cannot land while any commit carries it.

```
git rm --cached <big-file>
```

fatal: could not read Username for 'https://github.com': No such device or address

Git wanted to ask for credentials but no terminal was attached, the classic CI and script failure.

```
git remote set-url origin https://x-access-token:${TOKEN}@github.com/<owner>/<repo>.git
```

fatal: The current branch has no upstream branch

This local branch has never been connected to a remote branch, so plain git push does not know where to go.

```
git push -u origin <branch>
```

nothing to commit, working tree clean

Commit found no staged changes. Either everything is already committed, or the edits live elsewhere.

```
git status
```



Errors, continued

error: cannot pull with rebase: You have unstaged changes

A rebase must start from a clean tree, and yours has edits.

```
git pull --rebase --autostash
```

fatal: the remote end hung up unexpectedly

The connection died mid-transfer, and Git may report it as early EOF. Usually a large clone or push meeting a proxy, server, or network limit.

```
git config http.postBuffer 157286400
```

remote: Repository not found.

The URL has a typo, the repository is private and you lack access, or you are authenticated as the wrong account.

```
git remote -v
```

Host key verification failed.

SSH does not yet trust this server's fingerprint, or the stored one no longer matches.

```
ssh -T git@github.com
```

warning: LF will be replaced by CRLF the next time Git touches it

Windows line endings meeting Unix line endings. Harmless, but configure it once and it goes quiet.

```
printf '* text=auto\n' >> .gitattributes
```

error: unable to create file <path>: Filename too long

Windows' 260-character path limit, hit by deep folder structures.

```
git config --global core.longpaths true
```

Everything up-to-date

Not an error: push found nothing to send. The surprise usually means the work is uncommitted or on another branch.

```
git status
```

error: You need to resolve your current index first

A merge stopped at conflicts, and Git will not move on until you finish or abandon it.

```
git status
```



The living version

Eight doors, one address: amey-thakur.github.io/GIT-GUIDE

Finder

ask anything, or paste
the error

Start

zero to first push

Learn

the model, step by step

Fix

guided rescue

Errors

Git's messages,
decoded

GitHub

first repo to branch
protection

Workflows

how teams run Git

Cheatsheet

everything, one line
each

SAFE**REWRITES HISTORY****DESTRUCTIVE**

every answer carries its danger level and its undo

Every answer also lives in the repository's Discussions; a question the guide cannot answer becomes its next addition. MIT licensed, no trackers, works offline.

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Every Git and GitHub answer in one place.